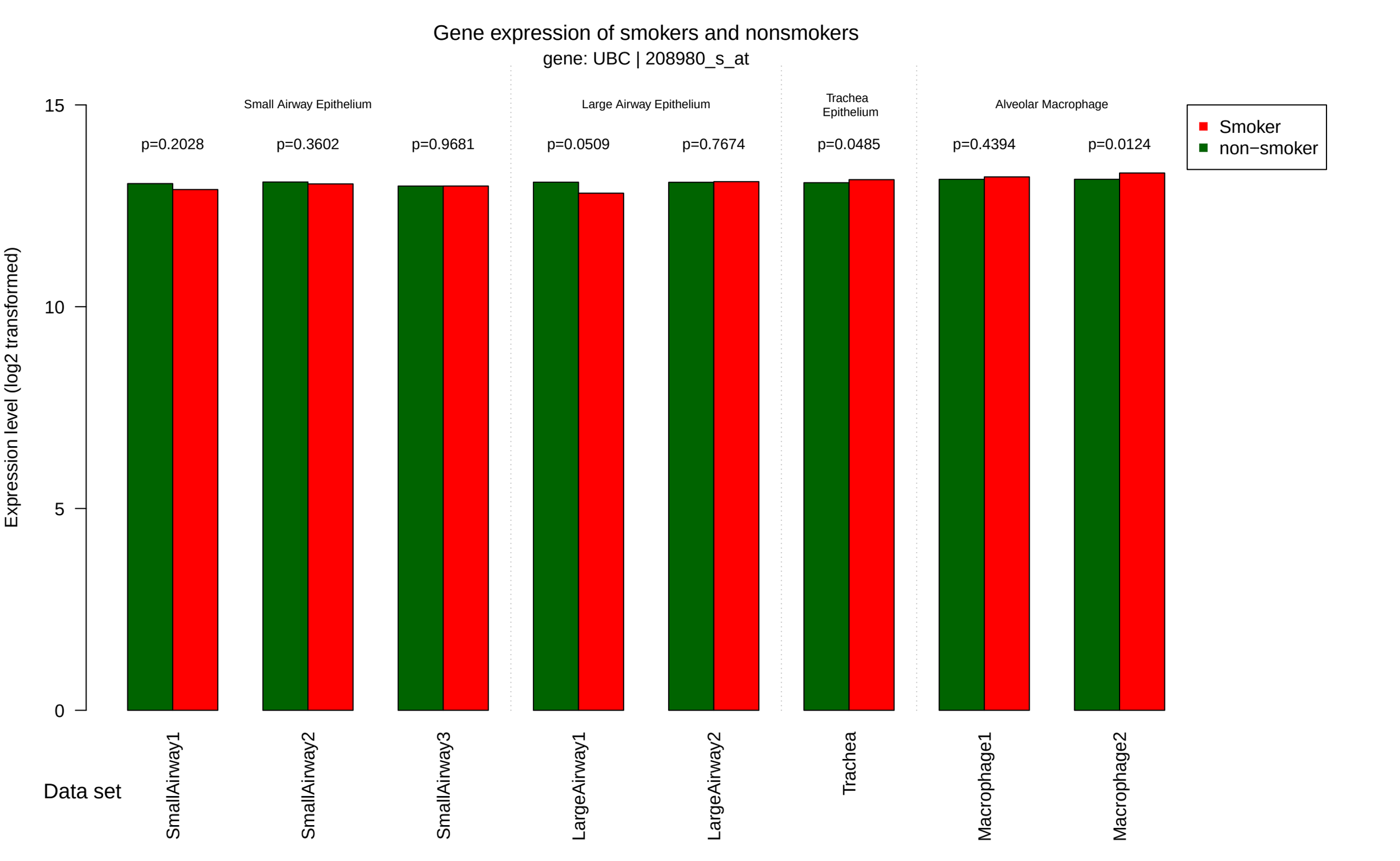
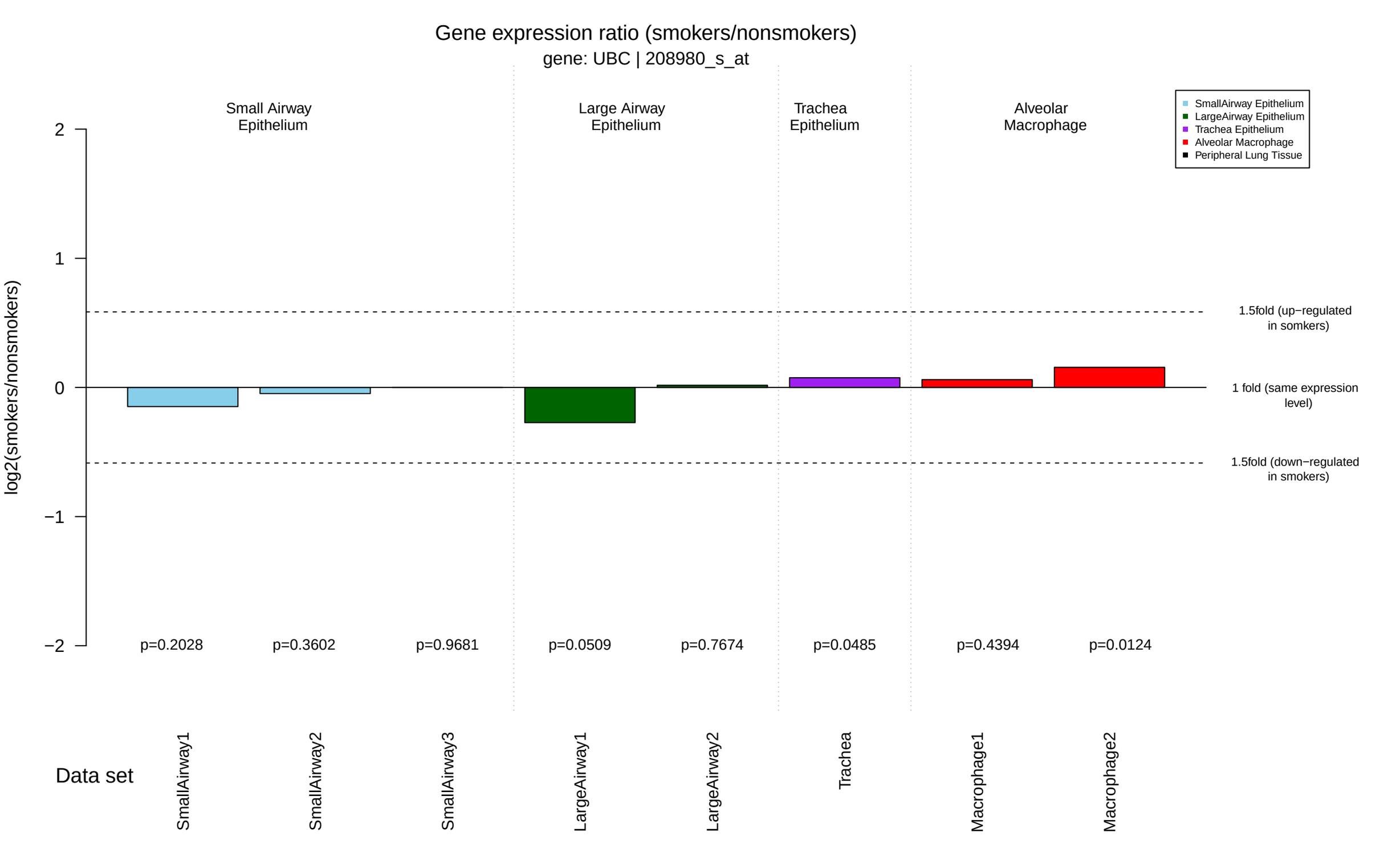
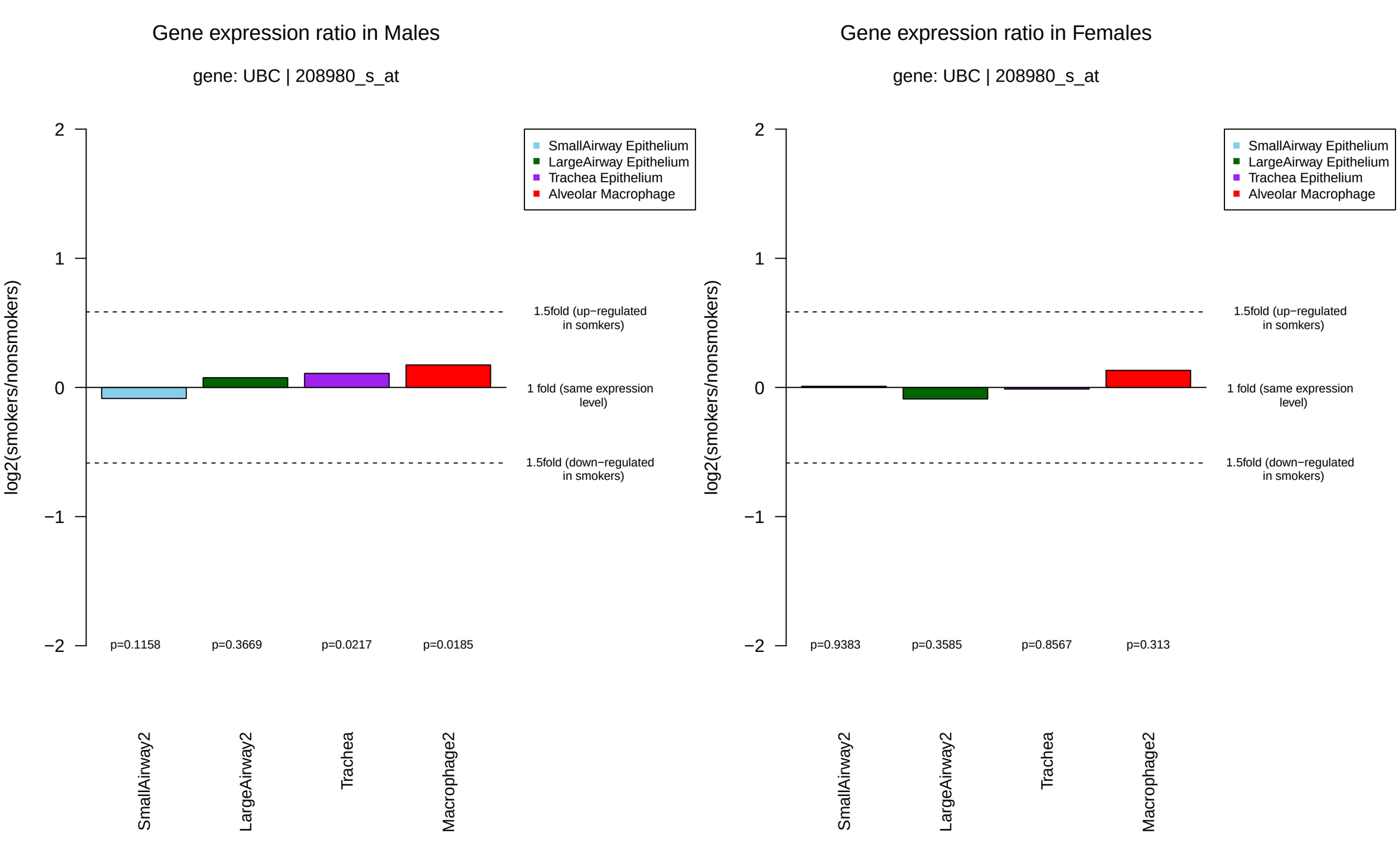
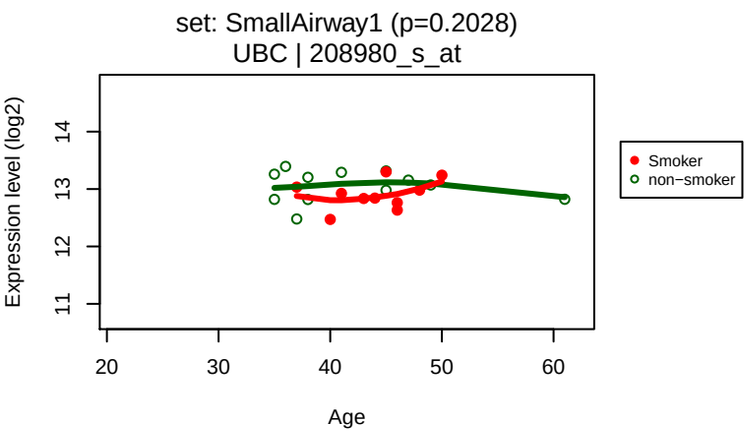
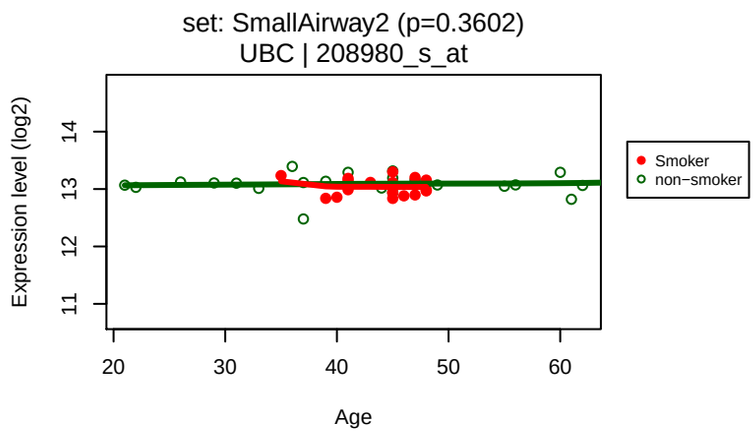
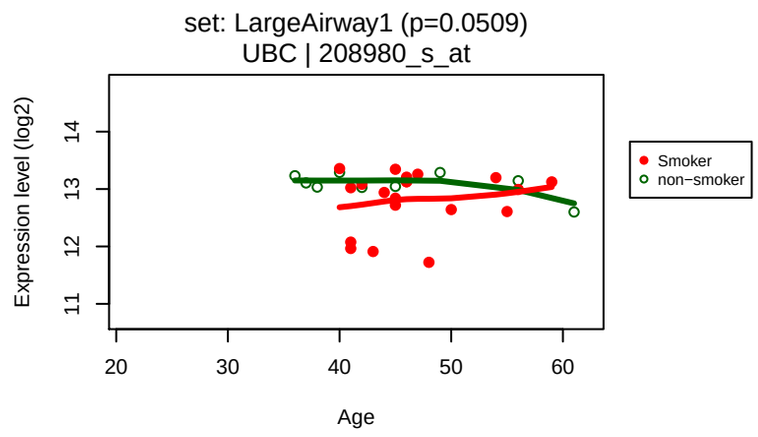

Gene expression of smokers and nonsmokers
gene: UBC | 208980_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2028
p=0.3602
p=0.9681
p=0.0509
p=0.7674
p=0.0485
p=0.4394
p=0.0124
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

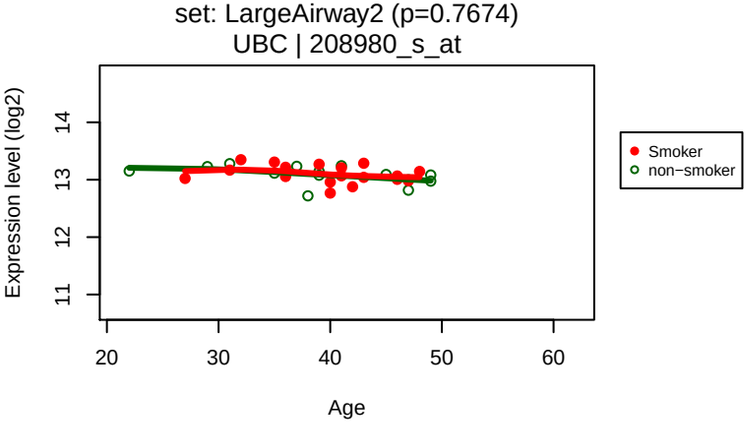

Gene expression ratio (smokers/nonsmokers)
gene: UBC | 208980_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2028
p=0.3602
p=0.9681
p=0.0509
p=0.7674
p=0.0485
p=0.4394
p=0.0124
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

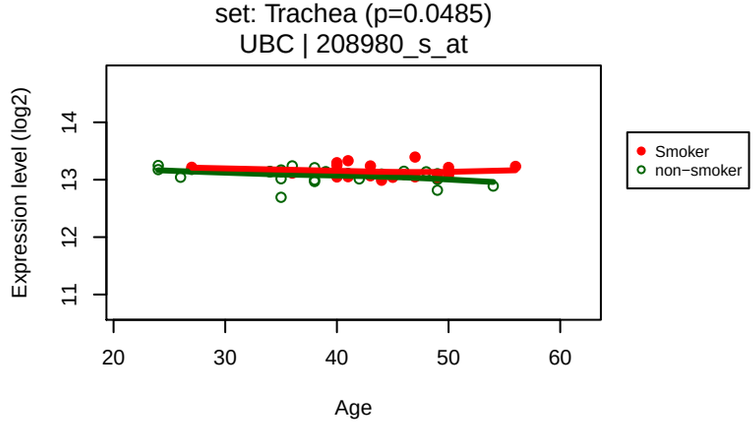

Gene expression ratio in Males
gene: UBC | 208980_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1158
p=0.3669
p=0.0217
p=0.0185
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: UBC | 208980_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9383
p=0.3585
p=0.8567
p=0.313
SmallAirway2
LargeAirway2
Trachea
Macrophage2

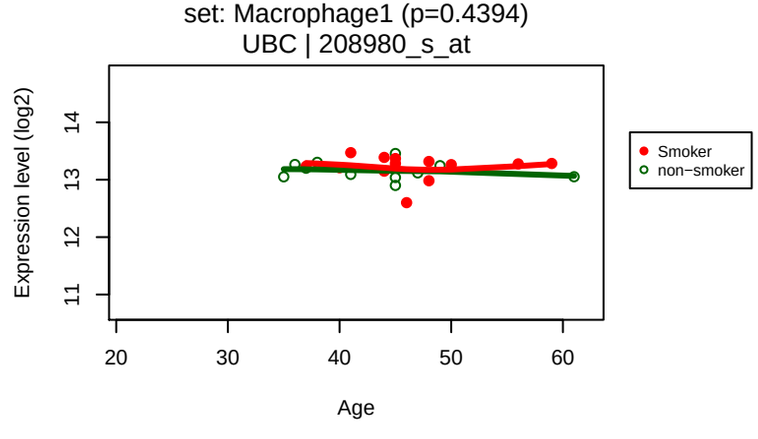

set: SmallAirway1 (p=0.2028)
UBC | 208980_s_at
Expression level (log2)
11
12
13
14
20
30
40
50
60
Age
Smoker
non−smoker

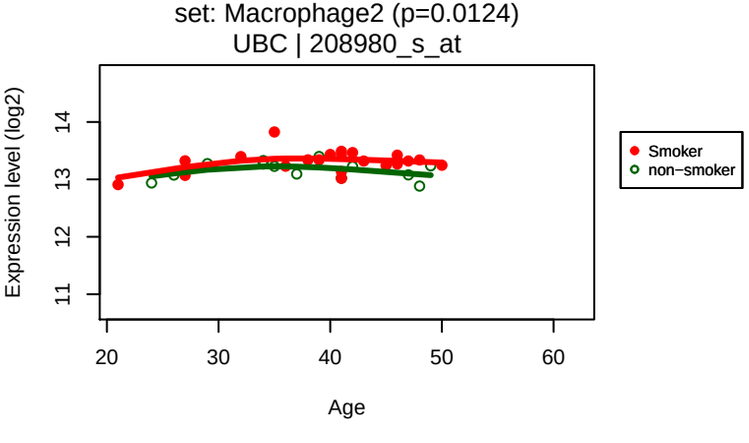

set: SmallAirway2 (p=0.3602)
UBC | 208980_s_at
Expression level (log2)
11
12
13
14
20
30
40
50
60
Age
Smoker
non−smoker

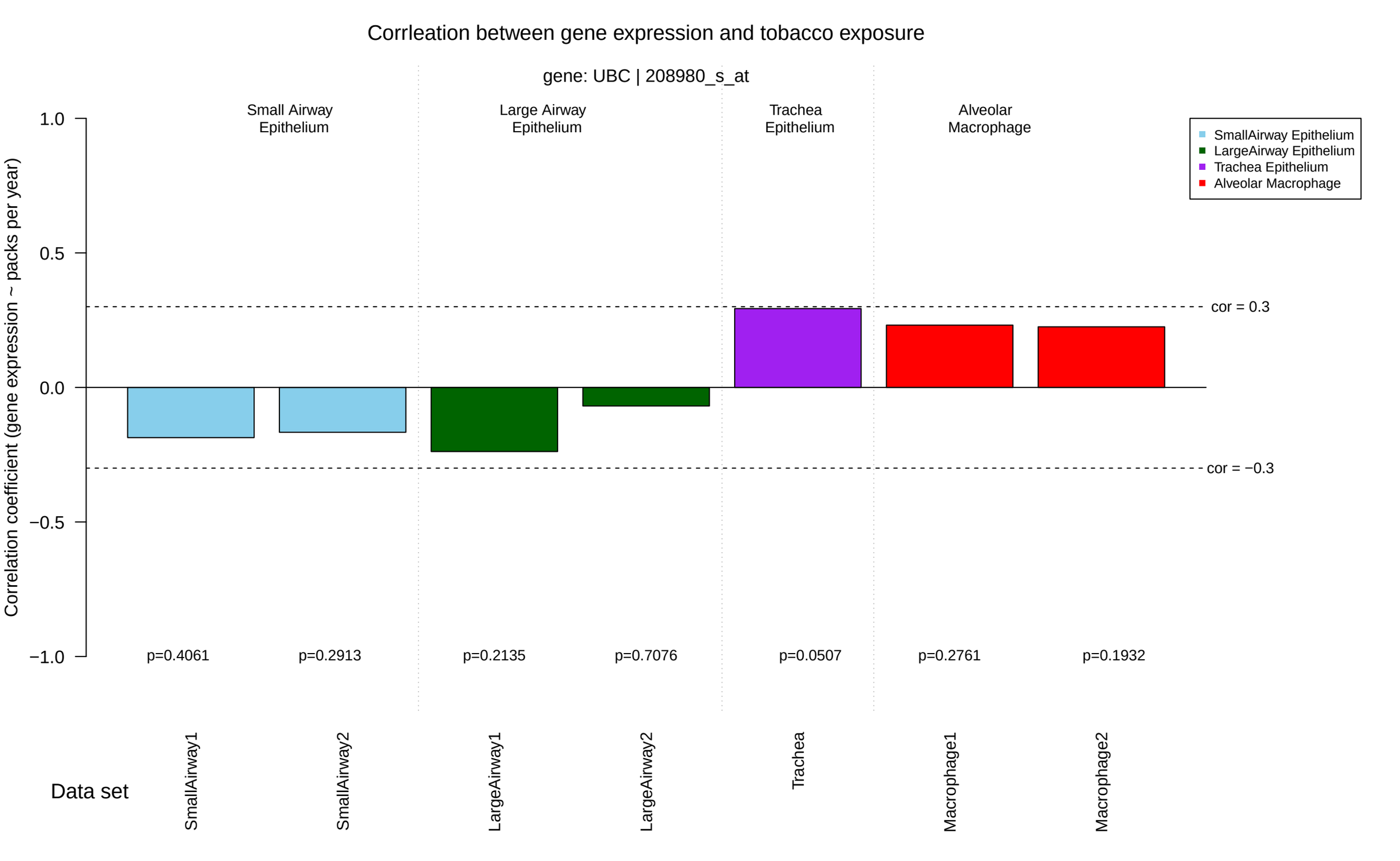

set: LargeAirway1 (p=0.0509)
UBC | 208980_s_at
Expression level (log2)
11
12
13
14
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7674)
UBC | 208980_s_at
Expression level (log2)
11
12
13
14
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0485)
UBC | 208980_s_at
Expression level (log2)
11
12
13
14
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4394)
UBC | 208980_s_at
Expression level (log2)
11
12
13
14
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0124)
UBC | 208980_s_at
Expression level (log2)
11
12
13
14
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: UBC | 208980_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.4061
p=0.2913
p=0.2135
p=0.7076
p=0.0507
p=0.2761
p=0.1932
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2